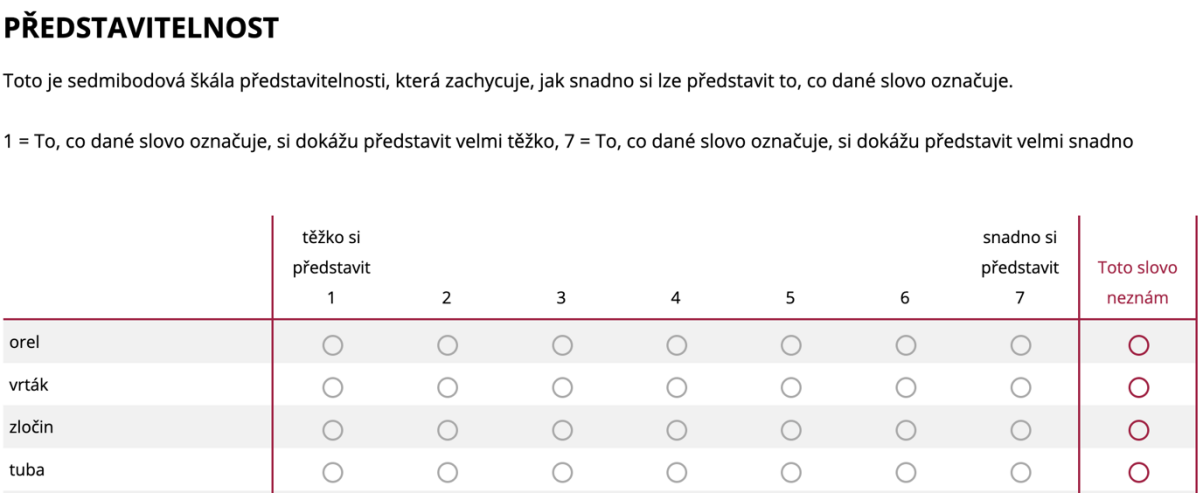
Table 1. *List of languages with at least one human derived large-scale norming database for either AoA, concreteness, imageability, valence or gender. The number of stimuli items being rated and the number of participants (ppts) is given, in addition to the citation. Where multiple sources are available for a given language, we refer to the paper with the largest number of items. If a dimension only has a smaller database available in a language, we also include that information, even if would not be considered large-scale.*

Language	AoA	Concreteness	Gender	Imageability	Valence
Arabic	items = 2,467, ppts = 803 Alzahrani et al. (2025)	items = 2,467, ppts = 803 Alzahrani et al. (2025)	-	-	-
British Sign Language	items = 300, ppts = 33 Vinson et al. (2008)	items = 234, ppts = 15 Ortega et al. (2025)	-	-	-
Chinese (Simplified)	items = 19,716, ppts = 1,765 Xu et al. (2021)	items = 9,877, ppts = 1,140 Xu & Li (2020)	-	items = 1,100, ppts = 960 Yao et al. (2017)	items = 11,310, ppts = 1,444 Xu et al. (2022)
Chinese (Traditional)	items = 4,376, ppts = 20 Su et al. (2023)	items = 25,281, ppts = 1,080 Chan & Tse (2024)	-	items = 25,281, ppts = 1,080 Chan & Tse (2024)	items = 25,281, ppts = 1,080 Chan & Tse (2024)
Croatian	items = 6000, ppts = 3630 Peti-Stantić et al. (2021)	items = 6,000, ppts = 3,630 Peti-Stantić et al. (2021)	-	items = 6,000, ppts = 3,630 Peti-Stantić et al. (2021)	items = 3,022, ppts = 933 Ćoso et al. (2019)
Czech	items = 299, ppts = 26 Łuniewska et al. (2019)	-	items = 3,999, ppts = 1,709 Preininger et al. (2025)	items = 435, ppts = 122 Smolík (2019)	items = 3,999, ppts = 1,709 Preininger et al. (2025)
Dutch	items = 30,000, ppts = 74 Brysbaert et al. (2014b)	items = 30,000, ppts = 75 Brysbaert et al. (2014b)	items = 24,037, ppts = 80 Vankrunkelsven et al. (2024)	items = 4,600, ppts = NA van Loon-Vervoom (1985)	items = 24,000, ppts = 94 Speed & Brysbaert (2024)
English	items = 30,121, ppts = 1,960 Kuperman et al. (2012)	items = 39,954, ppts = 4,000 Brysbaert et al. (2014a)	items = 5,553, ppts = 829 Scott et al. (2019)	items = 5,553, ppts = 829 Scott et al. (2019)	items = 13,915, ppts = 1,827 Warriner et al. (2013)
Estonian	-	items = 35,979, ppts = 2,278 Proos & Aigro (2024)	-	-	-
Finnish	items = 299, ppts = 24 Łuniewska et al. (2016)	-	-	-	items = 420, ppts = 996 Söderholm et al. (2013)
French	items = 1,493, ppts = 59 Ferrand et al. (2008)	items = 1,659, ppts = 482 Bonin et al. (2018)	-	items = 1,493, ppts = 30 Bonin et al. (2011)	items = 1,659, ppts = 482 Bonin et al. (2018)

French Sign Language	-	items = 546, ppts = 33 Périn et al. (2024)	-	-	-
German	items = 3,259, ppts = 537 Birchenough et al. (2017)	items = 1,000, ppts = 64 Kanske & Kotz (2010)	-	items = 2,900, ppts = 200 Võ et al. (2009)	items = 2,900, ppts = 200 Võ et al. (2009)
German Sign Language	items = 313, ppts = 62 Trettenbrein et al. (2021)	items = 234, ppts = 18 Ortega et al. (2025)	-	-	-
Indonesian	-	items = 1,490, ppts = 328 Sianipar et al. (2016)	-	-	items = 1,490, ppts = 328 Sianipar et al. (2016)
Italian	items = 1,957, ppts = 507 Montefinese et al. (2019)	items = 1,121, ppts = 400 Montefinese et al. (2014)	-	items = 1,121, ppts = 400 Montefinese et al. (2014)	items = 1,121, ppts = 684 Montefinese et al. (2014)
Persian	items = 299, ppts = 21 Łuniewska et al. (2019)	-	-	-	items = 718, ppts = 463 Mahjoubnavaz et al. (2024)
Polish	items = 4,900, ppts = 400 Imbir (2016)	items = 4,900, ppts = 400 Imbir (2016)	-	items = 4,900, ppts = 400 Imbir (2016)	items = 6,000, ppts = 21,878 Wierzba et al. (2022)
Portuguese (European)	items = 1,749, ppts = 685 Cameirao & Vicente (2010)	items = 3,800, ppts = 2,357 Soares et al. (2017)	-	items = 3,800, ppts = 2,357 Soares et al. (2017)	items = 1,034, ppts = 958 Soares et al. (2012)
Serbian	items = 2,100, ppts = 603 Popović Stijačić & Filipović Đurđević (2025)	items = 2,100, ppts = 603 Popović Stijačić & Filipović Đurđević (2025)	-	items = 2,100, ppts = 603 Popović Stijačić & Filipović Đurđević (2025)	items = 2,100, ppts = 603 Popović Stijačić & Filipović Đurđević (2025)
Spanish	items = 7039, ppts = 2,150 Alonso et al. (2015)	items = 1,400, ppts = 826 Guasch et al. (2016)	-	items = 1,400, ppts = 826 Guasch et al. (2016)	items = 14,031, ppts = 350 Stadthagen-Gonzalez et al. (2017)
Spanish (Rioplatense)	-	items = 1,034, ppts = 483 Sarli & Justel (2022)	-	items = 1,034, ppts = 483 Sarli & Justel (2022)	items = 1,034, ppts = 483 Sarli & Justel (2022)
Turkish	items = 503, ppts = 134 Conca et al. (2024)	items = 2,031, ppts = 1,527 Kapucu et al. (2021)	-	items = 503, ppts = 134 Conca et al. (2024)	items = 2,031, ppts = 1,527 Kapucu et al. (2021)

Table 2. *Dimensions that have been normed, with the types of scales used in the experiment and the anchors used for the scale. The calibrator and control words are also given for each of the dimensions, along with example items that have mean values across the span of the scale. English translation equivalents are given in square brackets.*

Dimension	Scale	Anchors	Calibrator/control	Examples
AoA	Free response numeric	NA	pejsek [*doggy*] federalismus [*federalism*]	mama [*mommy*] (1.63), bota [*shoe*] (3.03), plavčík [*lifeguard*] (6.58), evakuace [*evacuation*] (8.81), badatel [*researcher*] (10.19), patologický [*pathological*] (14.26), bitcoin [*bitcoin*] (18.06)
Concreteness	5-point Likert	1 = abstract 5 = concrete	teorie [*theory*] neoliberalismus [*neoliberalism*]	jestli [*if*] (1.1), rozum [*reason*] (1.88), úplně [*completely*] (2.20), jemný [*fine*] (3.14), francouzština [*French*] (3.79), kresba [*drawing*] (4.33), bunda [*jacket*] (5.0)
Gender	7-point Likert	-3 = very masculine 3 = very feminine	náhrdelník [*necklace*] otec [*father]*	varlata [*testicles*] (-2.71), mnich [*monk*] (-2.04), bořit [*to demolish*] (-0.94), sůl [*salt*] (0.01), červenat se [*to blush*] (1.08), zdravotnice [*nurse*] (2.06), paní [*lady*] (2.85)
Imageability	7-point Likert	1 = difficult to imagine 7 = easy to imagine	orel [*eagle*] chipsy [*crisps*]	však [*however*] (1.55), zda [*whether*] (2.03), možný [*possible*] (3.04), apatie [*apathy*] (3.97), autismus [*autism*] (5.00), komik [*comedian*] (6.05), brambor [*potato*] (7.00)
Valence	7-point Likert	-3 = very negative 3 = very positive	šikanovat [*to bully*] štestí [*happy*]	týrání [*abuse*] (-2.93), epidemie [*epidemic*] (-1.94), truc [*spite*] (-1.03), kapuce [*hood*] (0.02), mýdlo [*soap*] (0.99), radovat [*to rejoice*] (2.08), šťastný [*happy*] (2.88)



PŘEDSTAVITELNOST

Toto je sedmibodová škála představitelnosti, která zachycuje, jak snadno si lze představit to, co dané slovo označuje.

1 = To, co dané slovo označuje, si dokážu představit velmi těžko, 7 = To, co dané slovo označuje, si dokážu představit velmi snadno

	těžko si představit 1	2	3	4	5	6	snadno si představit 7	Toto slovo neznám
orel	○	○	○	○	○	○	○	○
vrták	○	○	○	○	○	○	○	○
zločin	○	○	○	○	○	○	○	○
tuba	○	○	○	○	○	○	○	○

Figure 1. *Example screenshot of the experimental set up for the imageability rating part of the experiment. The words being rated are on the right-hand side, with the scale being anchored by 'těžko si představit' [difficult to imagine] and 'snadno si představit' [easy to imagine]. The instructions are presented at the top of the screen, see the OSF repository for a full list of instructions along with their English translations.*

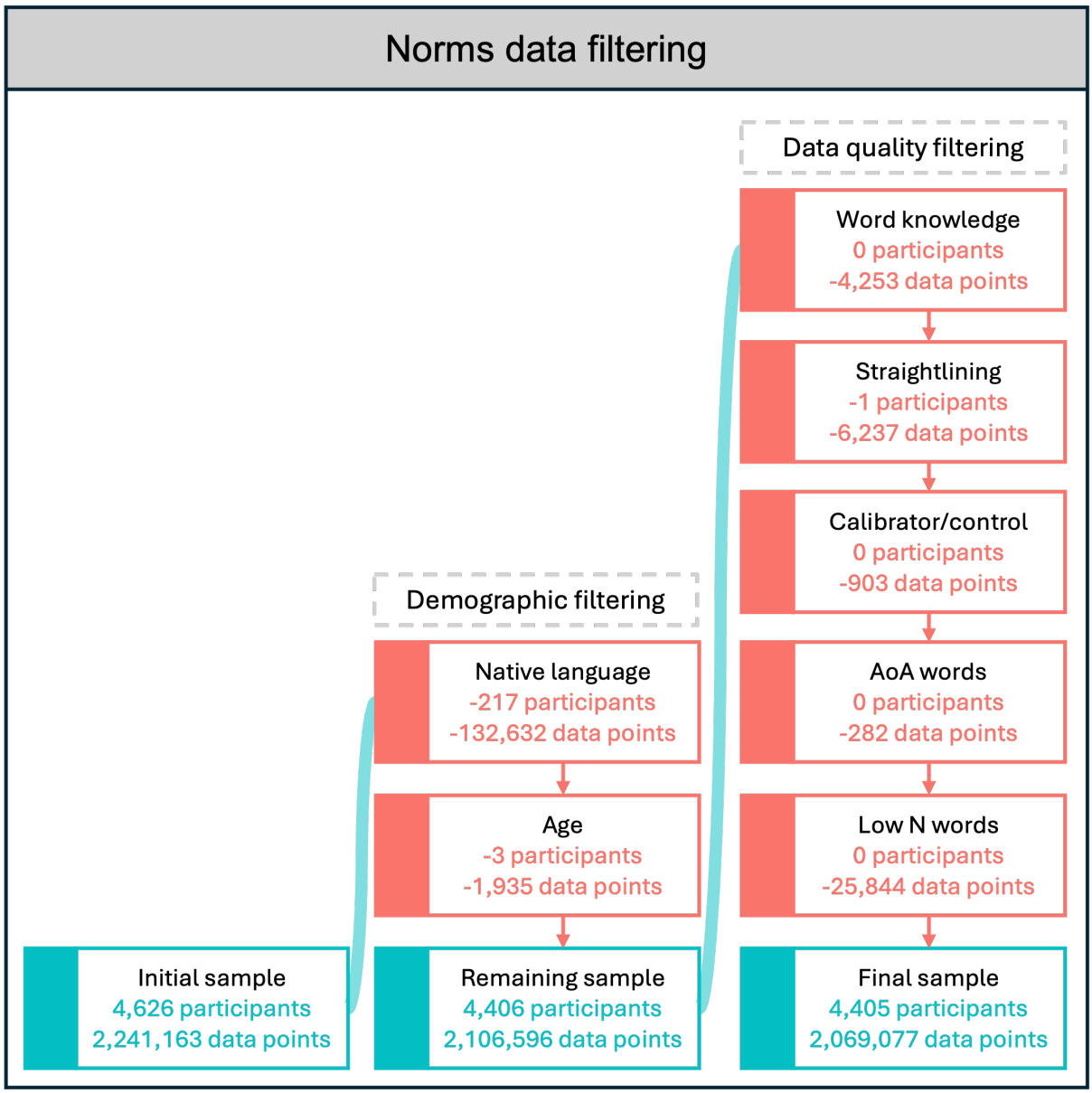


Figure 2. *Filtering procedure used on the raw data. Turquoise boxes contain summaries of the data that remain after each filtering block; red boxes contain summaries of the data removed during each filtering step.*

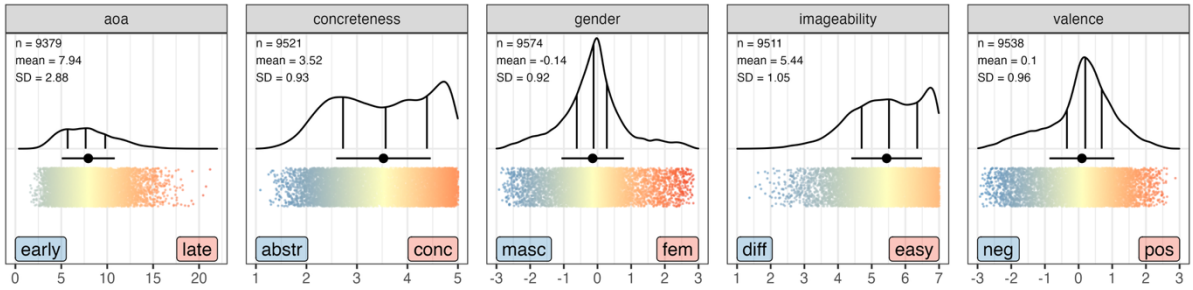


Figure 3. *Distribution of mean ratings for the AoA, concreteness, gender, imageability and valence dimensions. Kernel density estimates are shown with 25%, 50% and 75% quantiles marked by solid vertical lines. The black point with a horizontal line represents the mean and standard deviation of the ratings. Each word is represented by a coloured point, with more red/blue colours indicating a stronger association towards a specific side of the scale, and yellow points associating more with the mean across all ratings.*

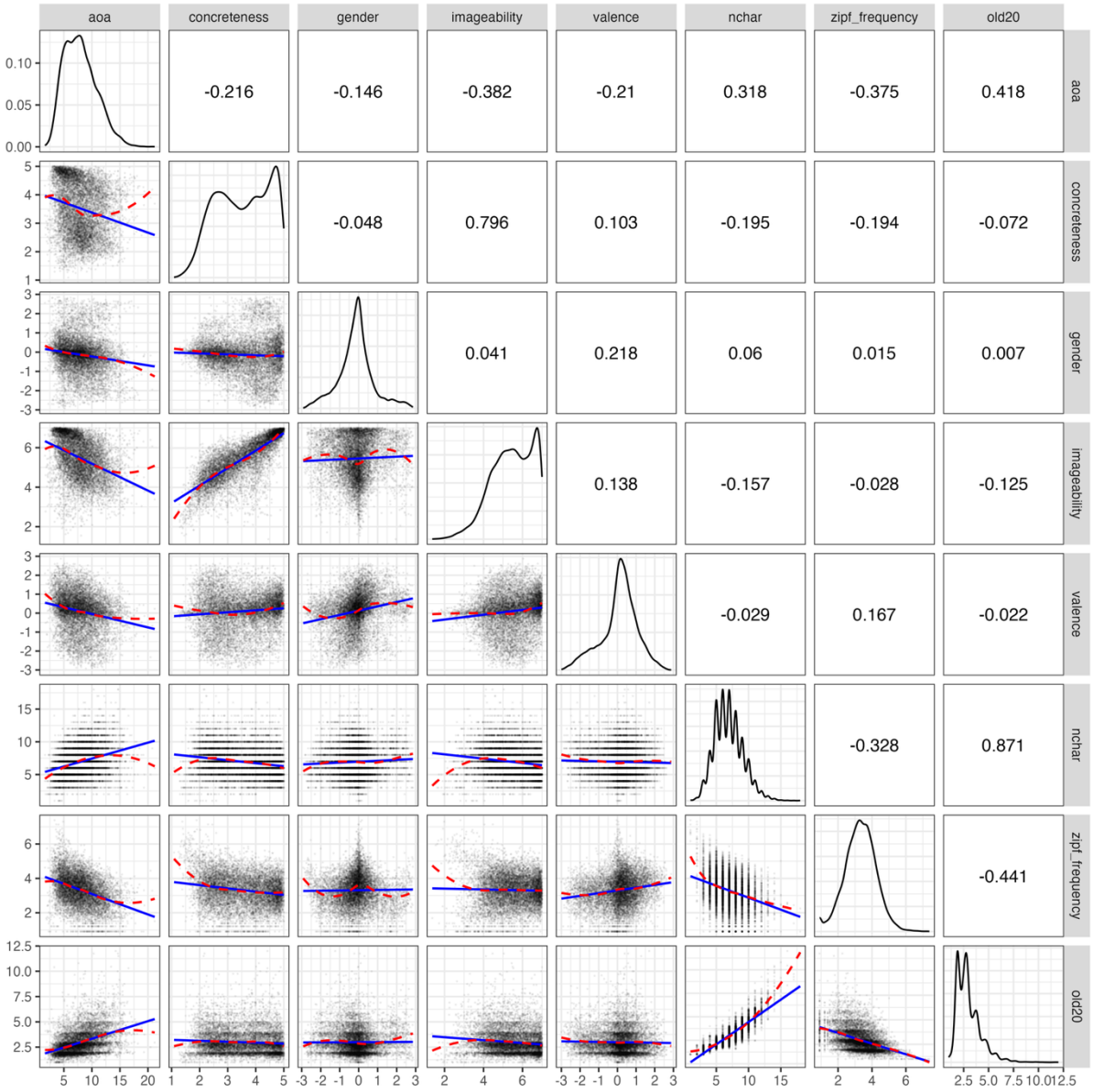


Figure 4. *Correlation matrix of all the dimensions in the word ratings dataset. The lower left plots are scatter plots of the data fitted with a linear (blue) and loess (red) fit, with the data from the dimension in the top facets on the x-axis and the data from the right facet on the y-axis. The diagonal plots show kernel density distributions of the variables. The upper right plots give the Pearson correlation coefficients between two dimensions.*

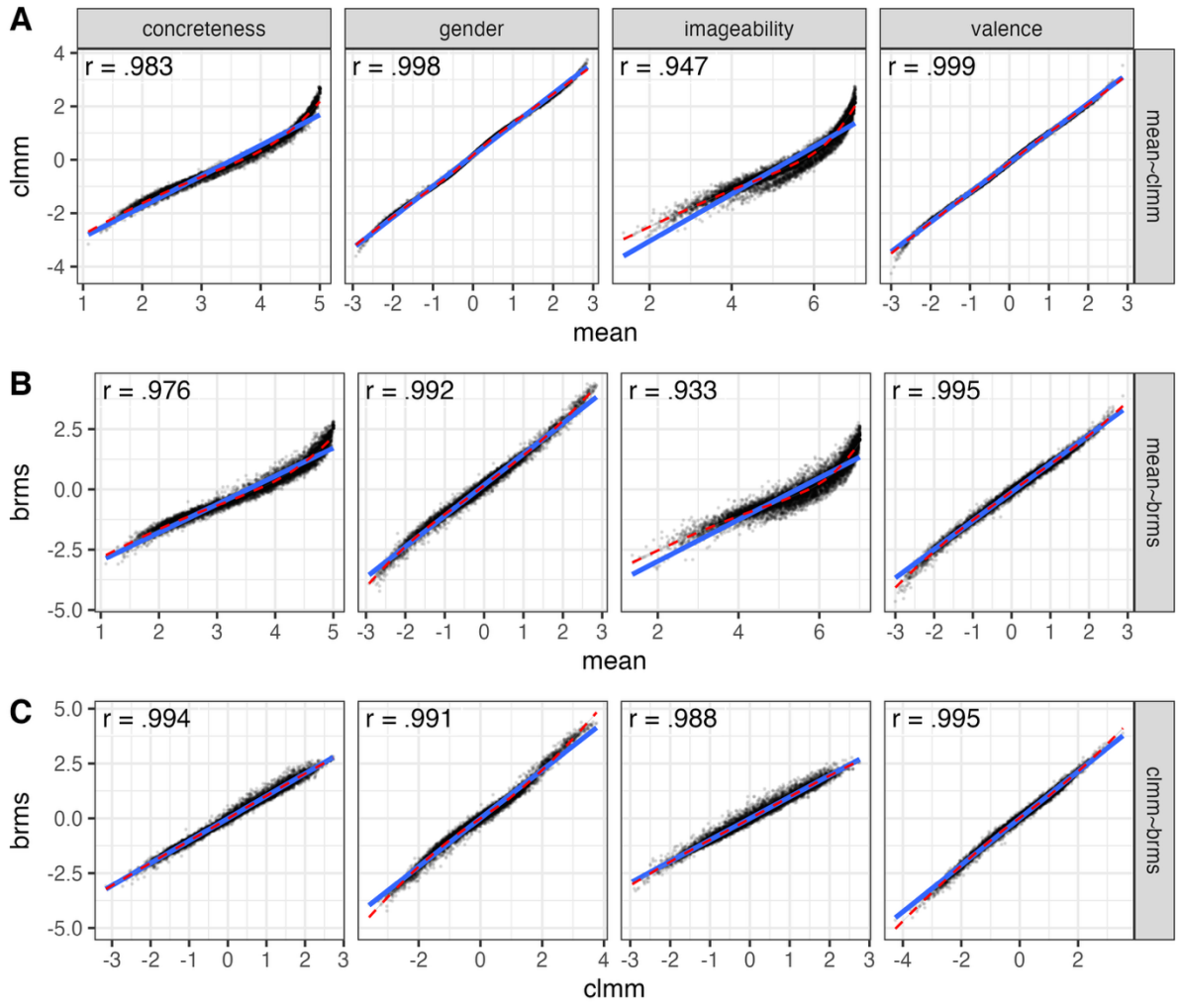


Figure 5. *Relationships between the three different aggregation methods across 4 different dimensions (top facets), with **A** giving the traditional mean on the x-axis and the clmm estimates on the y-axis, **B** giving the traditional mean on the x-axis and the brms estimates on the y-axis, and **C** giving the clmm estimates on the x-axis and the brms estimates on the y-axis. The blue lines provide a linear fit to the data, and the red dashed lines a loess fit. The correlation coefficients are given in the top left corner of each sub-plot.*

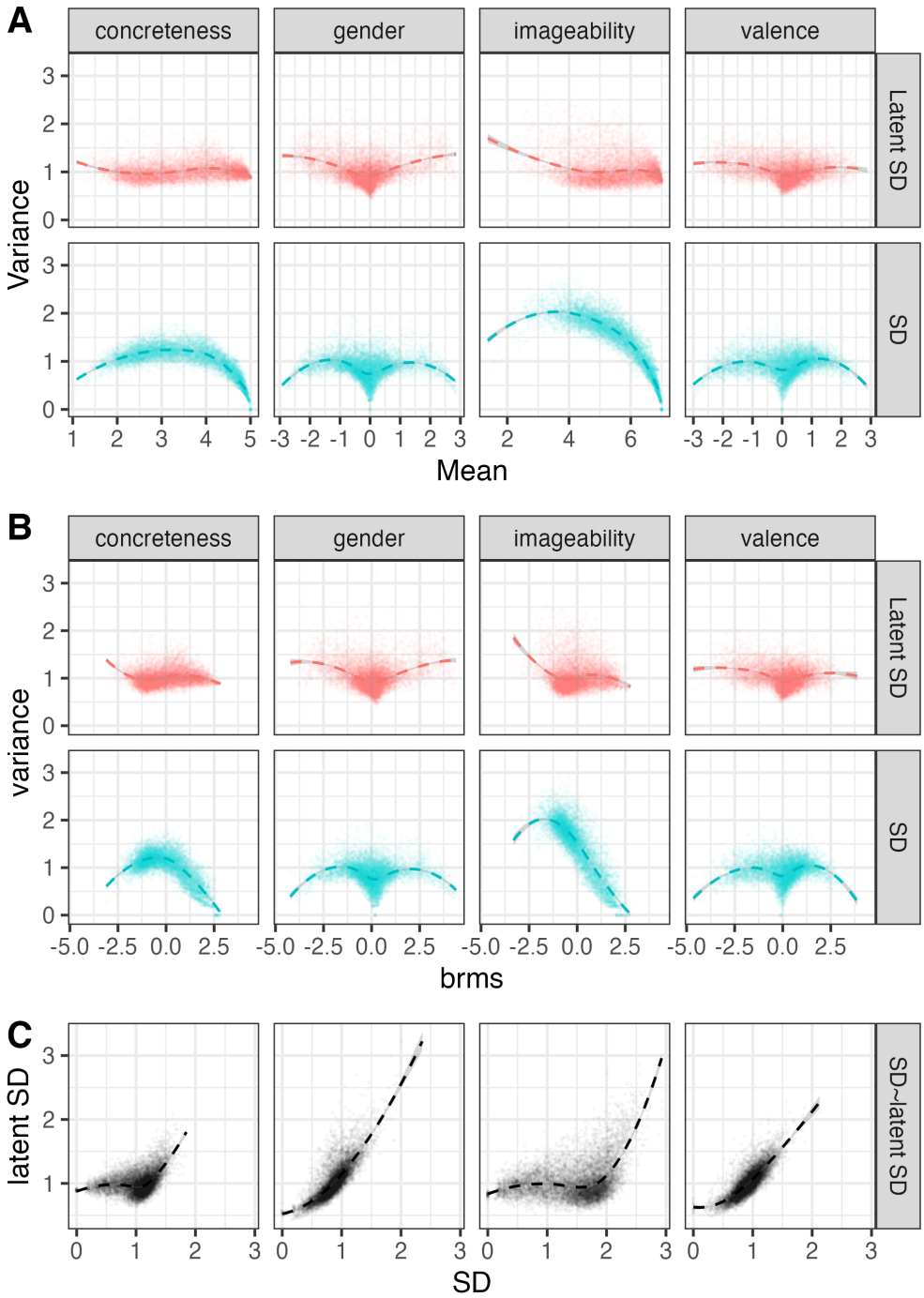


Figure 6. *Relationships between the different measures of variance across 4 different dimensions (top facets), with **A** giving the traditional mean on the x-axis and the variance based on the traditional SD and latent SD on the y-axis. **B** giving the same relationships as A, but with the brms estimates on the x-axis. C gives the relationship between the traditional SD and the latent SD. All of the plots are fitted with loess smooths.*

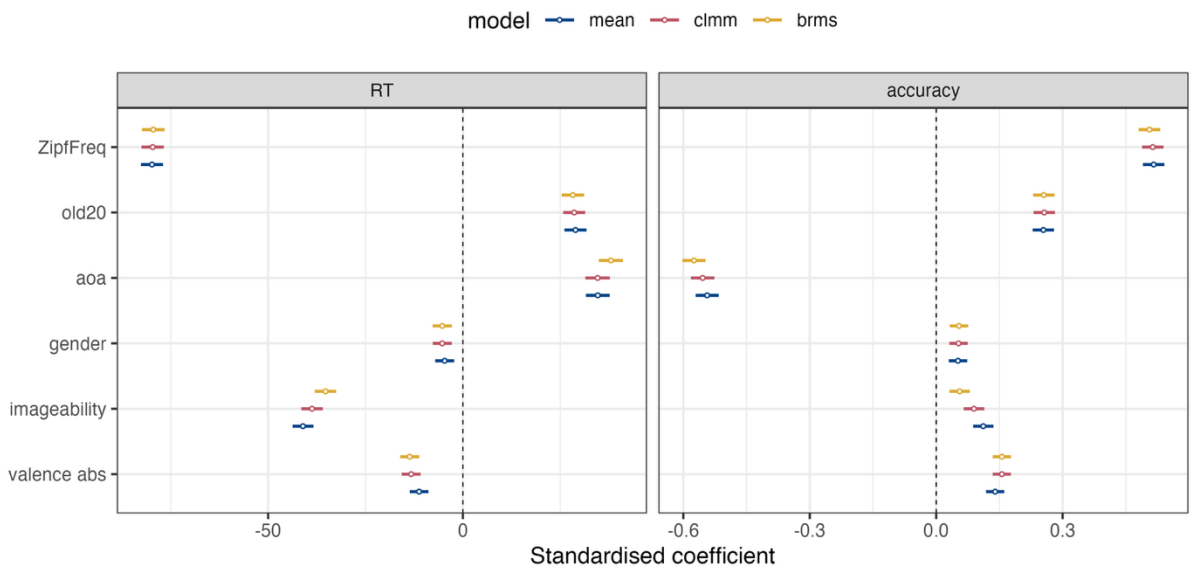


Figure 7. *Model coefficients and 95% confidence intervals when predicting data from the lexical decision task. The x-axis provides the standarised coefficients, with the predictor variables given on the y-axis. The left facet visualises RTs and the right facet accuracy. The colours represent the different models from which the values for each predictor variable was derived (blue = mean, red = clmm and yellow = brms).*